

T-Test and Group Comparisons

Statistical Methods Dossier with Subject-Specific Examples

1. Method Description

The t-test compares the means of two groups to determine if they differ significantly.

Types: (1) One-sample t-test: compares sample mean to a known population mean.

(2) Independent t-test: compares means of two independent groups.

(3) Paired t-test: compares two measurements from the same subjects (pre/post).

Test statistic: $t = (\text{mean1} - \text{mean2}) / \text{SE_diff}$. Degrees of freedom vary by test type.

Welch's t-test does not assume equal variances (robust alternative).

2. Subject-Specific Examples

Social Sciences (independent): Do men and women differ in income? Groups: male/female. DV: monthly income (EUR).

Economics (paired): Did a training program increase productivity? Pre vs. post scores.

Human Sciences (independent): Does a new drug reduce blood pressure more than placebo? Groups: drug/placebo. DV: systolic BP reduction.

Humanities (one-sample): Is the average sentence length of author A different from the historical average of 18.5 words?

3. Important Concepts and Assumptions

Normality: DV should be approximately normally distributed in each group (Shapiro-Wilk, Q-Q plot).

Independence: Observations must be independent (except paired t-test).

Homogeneity of variances: Variances should be equal across groups (Levenes test). If violated, use Welch.

No outliers: Extreme values can bias the t-test. Check boxplots.

4. Interpretation and practice

Report: 'An independent-samples t-test showed a significant difference between groups, $t(78) = 2.34$, $p = .022$, $d = 0.53$.'

Effect size (Cohen's d): $d = (M1 - M2) / \text{SD_pooled}$. Benchmarks: 0.2 (small), 0.5 (medium), 0.8 (large).

Confidence interval for the mean difference should always be reported.

One-tailed vs. two-tailed tests: Two-tailed is default. One-tailed requires strong directional hypothesis.

5. Code Examples

R (independent): `t.test(y ~ group, data = d, var.equal = TRUE); wilcox.test(y ~ group) # nonparametric`

R (paired): `t.test(pre, post, paired = TRUE)`

Python: `from scipy import stats; stats.ttest_ind(group1, group2); stats.ttest_rel(pre, post)`

SPSS: T-TEST GROUPS=group(0 1) /VARIABLES=y /CRITERIA=CI(.95).